# Childhood Obesity

26

## Introduction

Childhood obesity has a significant short- and long-term impact. The key to management lies in avoiding unnecessary work-up with physiological causes while not missing pathological causes.

## Diagnostic Criteria

- ☑ *Body mass index (after 2 years of age):* Weight (kg)/height ($m^2$)
  - *IAP 2015 charts* derived from backtracking of adult body mass index (BMI) Indian cutoffs for overweight (23 kg/$m^2$) and obesity (27 kg/$m^2$).
  - *Extreme obesity:* BMI ≥ 120% of the 95$^{th}$ percentile or ≥ 35 kg/$m^2$.
  - *Limitations:*
    - Ethnic differences in the proportion of body fat at the same BMI.
    - Lower reliability in pubertal age (pubertal children and adolescents are more likely to be classified as obese than those who are of the same age but prepubertal).
    - Erroneous diagnosis of obesity in short and muscular individuals.
- ☑ *Weight for height (weight for length):* Till 2 years of age (>97.7th percentile suggest obesity).

## Causes

Majority (~98%) do not have a pathological cause (exogenous or constitutional obesity).

- ☑ *Exogenous obesity:* Normal growth, development, and puberty are suggestive.
- ☑ *Monogenic obesity:* Early-onset obesity (before 5 years of age) with extreme hyperphagia (food-seeking behavior-stealing food, eating food leftover by others; impaired satiety).
- ☑ *Obesity syndromes:* Distinct features (abnormal facies, digits, vision) and systemic involvement, with hyperphagia.
- ☑ *Hypothalamic obesity:* Neurological features (headache, irritability, seizures) and/or neurological insult with rapid weight gain, hyperphagia. Neuroimaging is essential to identify a hypothalamic lesion.
- ☑ *Drug-induced obesity:* Associated with glucocorticoids, antipsychotics (risperidone and olanzapine), and antiepileptic drugs (valproate and carbamazepine).
- ☑ *Endocrine causes (hypothyroidism, Cushing syndrome, pseudohypoparathyroidism):* Commonly over-diagnosed due to confounding effects, but otherwise rare. Associated short stature is the hallmark of underlying endocrine cause. Obesity causes mild elevation of thyroid-stimulating hormone (TSH) that is usually its effect and not the cause.

## Assessment

Key aspects include differentiation of physiological and pathological causes, identification of a cause, and assessment of complications.

### Is it Pathological?

Any of the following suggest a pathological cause:

- ☑ Delay in growth, puberty, or development
- ☑ Dysmorphism
- ☑ Hyperphagia
- ☑ Early-onset
- ☑ Visual symptoms
- ☑ Neurological features.

**What is the Cause?**

- ☑ Birth weight, lifestyle, dietary intake, screen time, and physical activity.
- ☑ Family history of obesity, hypertension, and dyslipidemia.
- ☑ Onset and progression.
- ☑ Pubertal staging (including measurement of stretched penile length in boys).
- ☑ Treatment history for drugs causing obesity.
- ☑ Assessment of pointers and features of syndromic obesity **(Tables 1 and 2)**.
- ☑ *Waist circumference:* Marker of abdominal (regional) adiposity and a risk factor for metabolic syndrome.

**TABLE 1:** Pointers to cause of obesity.

| ***Features*** | ***Likely disorders*** |
|---|---|
| Delayed puberty | Bardet–Biedl, Prader–Willi |
| Retinitis pigmentosa, polydactyly | Alstrom, Bardet–Biedl |
| Short hands and feet | Prader–Willi syndrome |
| Buffalo hump, striae, plethora, hypertension | Cushing syndrome |
| Short fourth metacarpal | Pseudohypoparathyroidism |
| Developmental delay | Prader–Willi, hypothyroidism, pseudohypo-parathyroidism (PHP) |

**TABLE 2:** Features of common causes of syndromic obesity.

| ***Disorders*** | ***Features*** | ***Diagnostic test*** |
|---|---|---|
| Prader–Willi | ☑ Infantile hypotonia and failure to thrive followed by rapid weight gain after two years<br>☑ Almond-shaped eyes, prominent philtrum<br>☑ Acromicria, hypogonadism<br>☑ Hyperphagia, behavioral abnormalities | Methylation-sensitive polymerase chain reaction (PCR) for chromosome 15 (imprinting disorder) |
| Bardet–Biedl | ☑ Polydactyly, retinitis pigmentosa<br>☑ Developmental delay, polyphagia<br>☑ Renal abnormalities | Next generation sequencing (NGS) (Oligogenic chromosome 11) |
| Alstrom | Dilated cardiomyopathy, type 2 diabetes, progressive loss of vision and hearing | NGS (*ALMS1* gene mutation) |

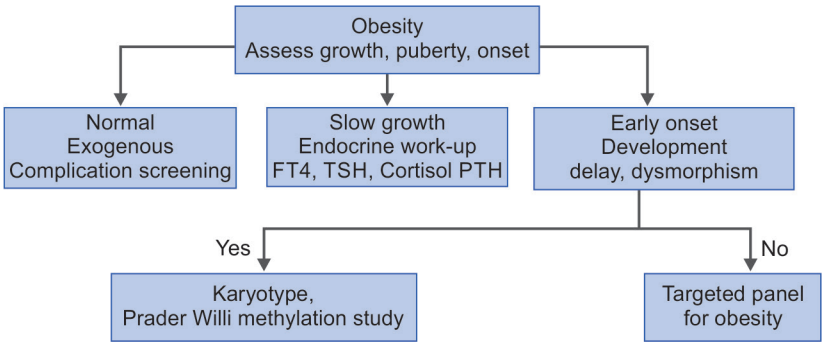
Assessment

**Pointers to Complications**

- ☑ Headache (benign intracranial hypertension, hypertension)
- ☑ Limp or leg pain (slipped capital femoral epiphysis)
- ☑ Day time somnolence (sleep apnea)
- ☑ Abdominal pain (gallstone disease and steatohepatitis) and polyuria (type 2 Diabetes).
- ☑ Cutaneous acanthosis.
- ☑ *Overdiagnosis:* Following conditions may be overdiagnosed in obese children.
  - Small phallic size due to buried penis.
  - Precocious puberty in girls with lipomastia.
  - Rickets in children with genu valgum.

**Work-up (Flowchart 1)**

**Flowchart 1:** Approach to obesity.




- ☑ *Complication assessment* ***(Table 3)****:* Key investigations include oral glucose tolerance test, ALT, and lipid profile. A sleep study may be considered with headache, daytime somnolence, and lethargy.
- ☑ *Endocrine work-up:* Only in children with short stature.
  - Morning cortisol and overnight dexamethasone suppression.
  - FT4, TSH.
  - Calcium, phosphorus, parathyroid hormone (PTH).
- ☑ *Targeted panel for monogenic causes:* Before 5 years with severe hyperphagia, delayed development.

## Assessment

**TABLE 3:** Pediatric cut-offs for metabolic complications.

| *Investigation* | *Level of concern* | *Pathological level* |
|---|---|---|
| Blood sugar fasting | 100–125 mg/dL | > 126 mg/dL |
| Blood sugar 2 hours after glucose* | 140–199 mg/dL | > 200 mg/dL |
| Hemoglobin A1c (HbA1c) | 5.7–6.4% | > 6.5% |
| Total cholesterol | 170–199 mg/dL | > 200 mg/dL |
| Low-density lipoprotein (LDL) cholesterol | 90–129 mg/dL | > 130 mg/dL |
| Triglyceride | 90–129 mg/dL | > 130 mg/dL |
| High-density lipoprotein (HDL) cholesterol | 40–45 mg/dL | < 40 mg/dL |
| Alanine aminotransferase (ALT) | > 25 IU/L (boys)<br>> 22 IU/L (girls) | > 60 IU/L |

*1.75 g/kg of glucose, to a maximum of 75 g—oral glucose tolerance test (OGTT).

## Prevention and Management

### Preventive Measures

- ☑ Exclusive breastfeeding till 6 months of age.
- ☑ Regular meal timings, including breakfast.
- ☑ At-least 7–8 hours of sleep daily at night.
- ☑ Lifestyle intervention should precede and should be maintained during pharmacotherapy.
- ☑ Obesity prevention guidelines from American Academy of Pediatrics recommend Fight Childhood Obesity by *'5-2-1-0' rule* **(Fig. 1)**. Accordingly, children can consume above 5 servings of fruits and vegetables, screen time below 2 hours, participate in one hour of physical activity, and consume 0 sugar-sweetened beverages daily.

**Fig. 1:** Prevention of childhood obesity "Let's Go: 5-2-1-0 Rule".

**Considerations for Specialist Referral**

- ☑ Early-onset obesity (before 5 years).
- ☑ Rapid progression.
- ☑ Delayed development, growth, and puberty.
- ☑ Neurological features
- ☑ Abnormal metabolic workup.



**Targets**

- ☑ Gradual and sustained loss.
- ☑ Body mass index (BMI) SD score (SDS) reduction of 5% roughly translating to 7–10% weight loss over 6 months.
- ☑ Avoid loss over 1.5 kg per month.

## Management

### General Measures

- ☑ Regular meals, including breakfast.
- ☑ 45 minutes of regular moderate to vigorous physical activity and screen time below 1 hour daily.
- ☑ Avoidance of snacking, inactivity, and screen exposure while eating.
- ☑ Avoid rigorous dieting.
- ☑ Identify red flag signs for psychosocial impact (school absenteeism, body shaming or teasing by peers regarding weight/appearance, persistent anxiety, depression or self-harm, anger outbursts, substance abuse, eating disorders) and family stressors.
- ☑ Specialist referral in the presence of complications.

### Medical Management

- ☑ *Metformin:* Approved in Type 2 DM after 8 years of age. May consider in causes related to antipsychotic medication, polycystic ovarian disease, and steatohepatitis.
- ☑ *Orlistat:* Gastric lipase inhibitor that decreases fat absorption. Approved after 12 years of age.

### Bariatric Surgery

- ☑ Should be discouraged as it carries more significant complications than adults.
- ☑ Indicated only with severe obesity (BMI >40 kg/m$^2$ or >35 kg/m$^2$ with complications) and only after completion of linear growth.
- ☑ A multidisciplinary obesity team with long-term follow-up is essential to maintain compliance with nutritional recommendations.
- ☑ Extreme motivation, strict diet, and activity schedule must be maintained after surgery.

## Further Reading


- ☑ Ahn SM. Current issues in bariatric surgery for adolescents with severe obesity: durability, complications, and timing of intervention. J Obes Metab Syndr. 2020;29(1):4-11.
- ☑ Barnett TA, Kelly AS, Young DR, Perry CK, Pratt CA, Edwards NM, et al. American Heart Association Obesity Committee of the Council on Lifestyle and Cardiometabolic Health; Council on Cardiovascular Disease in the Young; and Stroke Council. Sedentary behaviors in today's youth: approaches to the prevention and management of childhood obesity: A Scientific Statement From the American Heart Association. Circulation. 2018;138(11):e142-e159.
- ☑ Rogers VW, Hart PH, Motyka E, Rines EN, Vine J, Deatrick DA. Impact of Let's Go! 5-2-1-0: a community-based, multisetting childhood obesity prevention program. J Pediatr Psychol. 2013;38(9):1010-20.
- ☑ Styne DM, Arslanian SA, Connor EL, Farooqi IS, Murad MH, Silverstein JH, et al. Pediatric obesity-assessment, treatment, and prevention: An Endocrine Society Clinical Practice Guideline. J Clin Endocrinol Metab. 2017;102(3):709-57.

AQ:
Please check and verify yellow highlighted text for its correctness.

Indian Academy of Pediatrics (IAP)

# STANDARD TREATMENT GUIDELINES 2022

# Bronchiolitis


*Lead Author*
**Pallab Chatterjee**

*Co-Authors*
**Vineet Sehgal, Samir Shah**


**Under the Auspices of the IAP Action Plan 2022**

**Remesh Kumar R**
IAP President 2022

**Upendra Kinjawadekar**
IAP President-Elect 2022

**Piyush Gupta**
IAP President 2021

**Vineet Saxena**
IAP HSG 2022–2023

© Indian Academy of Pediatrics

## IAP Standard Treatment Guidelines Committee

***Chairperson***

**Remesh Kumar R**

***IAP Coordinator***

**Vineet Saxena**

***National Coordinators***

**SS Kamath, Vinod H Ratageri**

***Member Secretaries***

**Krishna Mohan R, Vishnu Mohan PT**

***Members***

**Santanu Deb, Surender Singh Bisht, Prashant Kariya, Narmada Ashok, Pawan Kalyan**